

TASK 2 REPORT: IMAGE ACQUISITION, REPRESENTATION AND IMAGE FUNDAMENTALS

- **Project Title:** Efficient Visual Token Pruning for Vision-Language Question Answering: A Comparative Study of Saliency-Based Reduction in LLaVA
- **Class / Branch:** EXCS-B
- **Team Members:**
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1. Aim of the Experiment

To develop the foundational image acquisition and representation module by:

1. Loading, displaying and verifying digital images as discrete multidimensional matrices.
2. Inspecting spatial coordinate dimensions, pixel data types and discrete intensity values.
3. Performing color space transformations across RGB, Grayscale and HSV domains.
4. Analyzing the visual and structural degradation caused by spatial decimation (sampling) and bit-depth reduction (amplitude quantization).
5. Benchmarking uncompressed (BMP), lossless (PNG) and lossy (JPEG) storage formats using quantitative compression ratios and distortion metrics.

2. Dataset & Sample Image Description

- **Dataset Source:** HuggingFaceM4/ChartQA Benchmark Dataset.
- **Domain:** Synthetic Document and Statistical Chart Image Processing.
- **Test Sample:** chart_01.png (selected from the project's curated dataset/images/ repository).
- **Sample Nature:** Multi-colored statistical bar chart with high-contrast text labels, grid lines and categorical color-coded bars over a uniform white canvas.

3. Task 2.3 & 2.4: Image Acquisition & Numerical Matrix Representation

Digital images in computer vision are represented as discrete 2D spatial coordinate grids mapped to intensity channels:

$$I(x, y) \in \{0, 1, \dots, 255\}^C$$

where x and y denote spatial coordinates and C denotes the channel depth.

Using OpenCV (cv2.imread), the input sample was acquired into memory and stored as a structured NumPy array.

Representation Metrics Extracted:

- **Dimensions ($H \times W$):** 600×850 pixels
- **Number of Channels:** 3 (Trichromatic color representation)
- **Total Image Resolution:** 510,000 spatial pixels (1,530,000 total byte elements)

- **Data Type:** uint8 (8-bit unsigned integer, dynamic range [0,255])
- **Queried Center Pixel** ($x = 425, y = 300$):
 - **OpenCV BGR Format:** $[B = 255, G = 255, R = 255]$
 - **Standard RGB Format:** $[R = 255, G = 255, B = 255]$

(Observation: The queried center pixel confirms a fully saturated white background, verifying that a significant proportion of the chart area consists of redundant non-informative whitespace.)

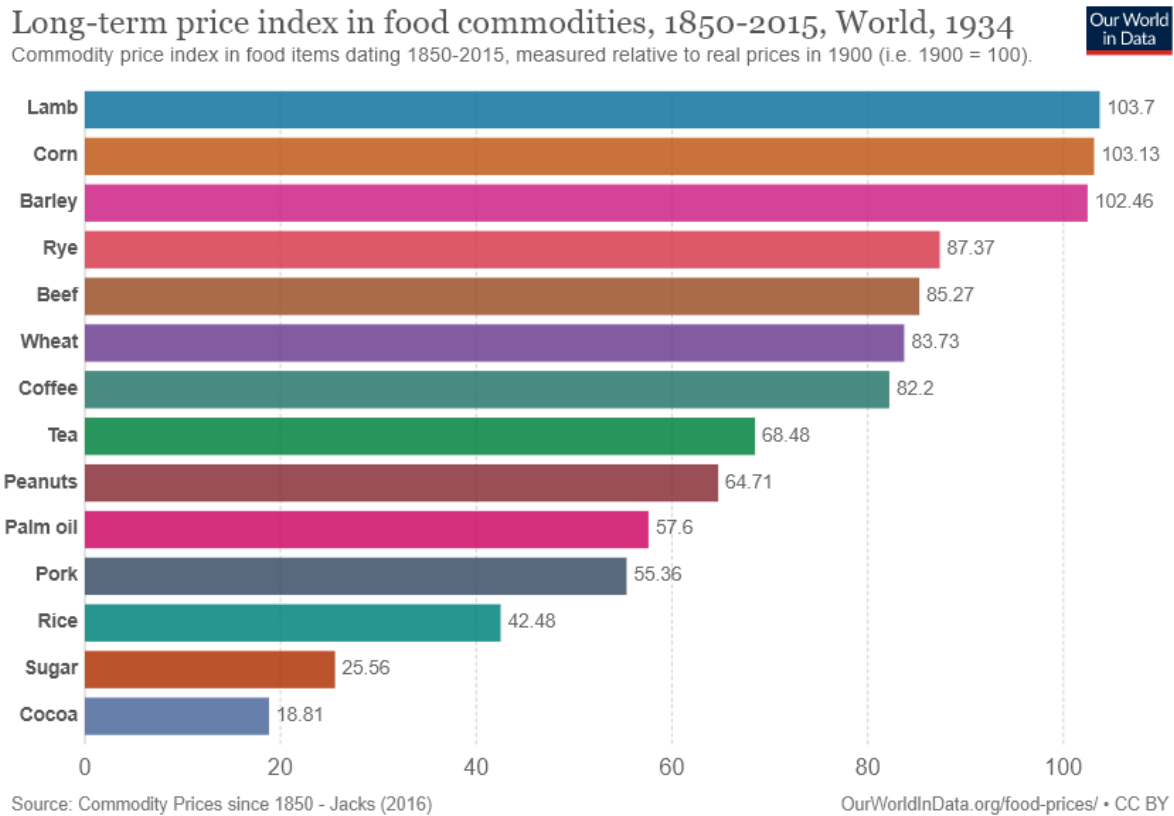


Figure 1: Acquired original chart image (chart_01.png) loaded at full resolution (600×850).

4. Task 2.5: Color Space Conversions & Analysis

The input chart was transformed into three distinct color models:

1. **RGB Color Space:** Standard additive display model. The image exhibits high correlation between color channels, making luminance and chromaticity difficult to separate.
2. **Grayscale Model:** Computed via standard luminance weighting:

$$Y = 0.299R + 0.587G + 0.114B$$

While suitable for edge detection, converting multi-colored charts to grayscale causes loss of categorical information when different color-coded bars share identical luminance intensities.

3. **HSV Color Space:** Decouples chromaticity into **Hue** (dominant wavelength, range $[0, 179]$ in OpenCV) and **Saturation** (color purity, range $[0, 255]$) from **Value** (brightness intensity, range $[0, 255]$). This decoupling allows reliable segmentation of chart bars by color regardless of shadows or brightness variations.

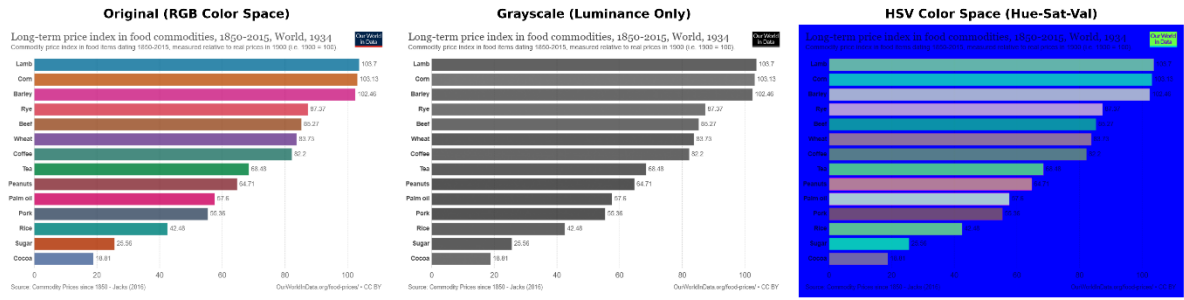


Figure 2: Visual comparison of Color Space Conversions: Original RGB (left), Grayscale Luminance (center) and HSV Representation (right).

5. Task 2.6: Sampling and Quantization Study

A. Spatial Sampling (Resizing & Resolution Degradation)

Spatial sampling discretizes the continuous visual field. The chart was evaluated across three spatial resolutions:

- **100% Sampling:** 600×850 pixels (Original spatial fidelity)
- **50% Sampling:** 300×425 pixels ($4 \times$ reduction in pixel count)
- **25% Sampling:** 150×212 pixels ($16 \times$ reduction in pixel count)

Observations:

- At **50% sampling**, major bar boundaries remain discernible, but small numerical annotations lose sharp definition.
- At **25% sampling**, severe spatial aliasing occurs. Fine textual labels, axis ticks and legend annotations blur into illegible pixel artifacts.
- **Connection to Final Project:** This observation demonstrates why feeding naively downsampled chart images into Vision-Language Models degrades Question Answering accuracy. It validates our project hypothesis: **selective token pruning at full resolution** preserves high-frequency textual details while naive spatial downsampling destroys them.

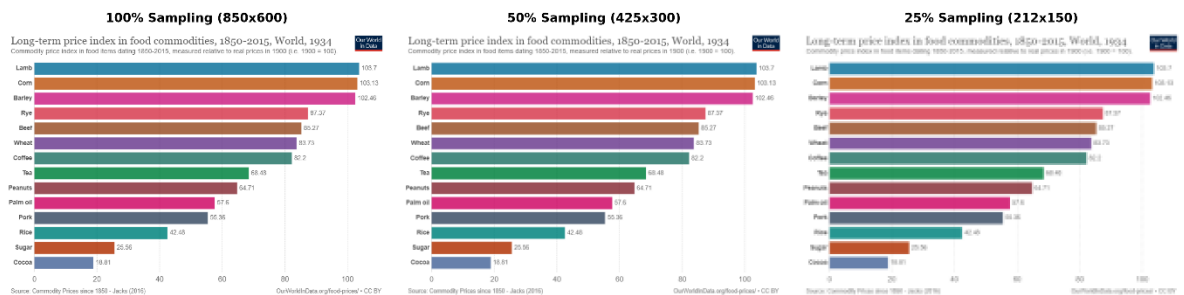


Figure 3: Spatial Sampling Analysis: 100% native resolution (left), 50% spatial decimation (center) and 25% aggressive downsampling displaying severe text aliasing (right).

B. Gray-Level Quantization (Bit-Depth Discretization)

Quantization discretizes the continuous luminance amplitude into $L = 2^k$ discrete gray levels:

$$I_Q(x, y) = \text{round}\left(\frac{I(x, y)}{\text{Step Size}}\right) \times \frac{255}{L - 1}, \quad \text{where Step Size} = \frac{256}{L}$$

The grayscale image was evaluated across 4 quantization bit-depths:

- **256 Levels (8-bit):** Standard continuous-tone representation.
- **128 Levels (7-bit):** Perceptually indistinguishable from 8-bit.
- **64 Levels (6-bit):** Subtle onset of false contouring around text antialiasing.
- **32 Levels (5-bit):** Pronounced **false contouring** (posterization). Continuous gray gradients separate into visible stair-stepped bands and character edges become blocky.

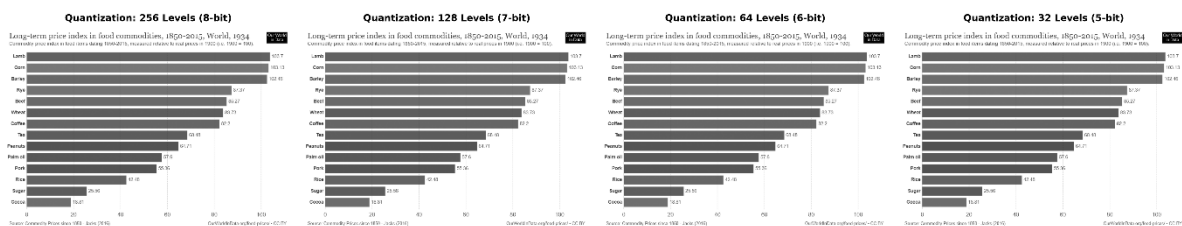


Figure 4: Amplitude Quantization comparison: 256 levels (8-bit), 128 levels (7-bit), 64 levels (6-bit) and 32 levels (5-bit) showing distinct false contouring bands.

6. Task 2.7: Image File Formats & Compression Benchmark

The uncompressed chart matrix was encoded into three standard storage container formats to benchmark compression efficiency and reconstruction fidelity:

Format	File Size (Bytes)	File Size (KB)	Compression Ratio	Compression Type	Visual Quality & Artifacts
BMP	1,531,254	1,495.37 KB	1.00× (Baseline)	None (Raw Raster Header + RGB)	Perfect pixel-for-pixel fidelity. Massive storage overhead.
PNG	37,631	36.75 KB	40.69×	Lossless (DEFLATE / LZ77 + Huffman)	Perfect mathematical fidelity; zero distortion.
JPEG	74,436	72.69 KB	20.57×	Lossy (8 × 8 Block DCT + Quantization, Q=85)	High-frequency ringing artifacts along sharp text and bar boundaries.

Quantitative Distortion Analysis:

- **Mean Squared Error (MSE):** 11.8278
- **Peak Signal-to-Noise Ratio (PSNR):** 37.40 dB

Critical Analytical Finding:

In natural scene photography, lossy JPEG usually produces much smaller files than PNG. However, for **synthetic chart documents, PNG achieved a file size 49% smaller than JPEG (36.75 KB vs. 72.69 KB) while remaining completely lossless.**

Reason: Synthetic charts consist of large contiguous regions of uniform pixel values (solid white background). PNG's spatial run-length and dictionary compressor (DEFLATE) compresses these uniform areas with near-zero entropy. Conversely, JPEG's 8×8 block Discrete Cosine Transform must encode high-frequency coefficients to approximate the step-function edges of chart text and gridlines, producing both higher file sizes and visible ringing artifacts.

7. Challenges Faced & Solutions Implemented

1. OpenCV BGR Channel Order:

- *Problem:* OpenCV defaults to BGR ordering, which causes red and blue components to invert when rendered via Matplotlib.
- *Solution:* Implemented explicit color channel re-ordering via `cv2.cvtColor(bgr_img, cv2.COLOR_BGR2RGB)` prior to visualization.

2. Dynamic Range Clipping During Quantization:

- *Problem:* Integer division when reducing gray levels produced values outside $[0, 255]$ during scaling, causing integer overflow.
- *Solution:* Normalized scaling using $(L - 1)$ step denominators combined with `np.clip(quantized, 0, 255).astype(np.uint8)`.

3. Distortion Quantification:

- *Problem:* Visual inspection alone cannot quantify lossy compression damage on text edges.
- *Solution:* Computed explicit mathematical MSE and PSNR metrics between the uncompressed bitmap and decompressed JPEG array.

8. Learning Outcomes

- Acquired practical mastery of discrete 2D image matrix structures, channel interleaving and data type ranges (uint8).
- Understood why HSV color space is better suited for color-based chart data segmentation than RGB.
- Confirmed through experiment that aggressive spatial downsampling destroys document legibility, justifying our project's approach of token pruning at full resolution over naive downscaling.
- Proved that lossless spatial encoding (PNG) is superior to lossy transform encoding (JPEG) for synthetic, text-dense chart imagery.